Runqi Wang

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Education

Shanghai Jiao Tong University Sep. 2023 – Jun. 2027 (Expected)

B.S. in Mathematics and Applied Mathematics & Artificial Intelligence Undergraduate, Rank : 2 / 37

Major Courses: Probability Theory, Mathematical Statistics, Stochastic Processes, Design and Analysis of Algorithms, Scientific Computing, Machine Learning, Deep Learning and Applications.

Research and Project Experience

Unlearning Offline Stochastic Multi-Armed Bandits Jul. 2025 – Jan. 2026

Research Co-first Author Submitted to NeurIPS 2026 (CCF-A)

- Studied machine unlearning for offline stochastic multi-armed bandits, focusing on decision-making under data deletion, privacy constraints, and uncertainty.
- Designed an adaptive algorithm combining **Gaussian mechanism** and **rollback** based on sample size and privacy budget; contributed to regret-style theoretical analysis and upper/lower bound proofs.
- Built large-scale simulation pipelines and validated the theoretical trade-off, achieving **20%–70%** expected performance improvement over baseline methods across multiple settings.

Efficient Stochastic Combinatorial Semi-Bandits Feb. 2025 – Jun. 2025

Research Second Author Submitted to ESA 2026 (CCF-B)

- Worked on stochastic combinatorial semi-bandit algorithms, with emphasis on efficient exploration, regret minimization, and scalable decision-making over large action spaces.
- Contributed to UCB-based algorithm design and theoretical analysis, deriving near-optimal regret guarantees under computationally efficient implementations.
- Led experiments on semi-bandit and cascading-bandit environments; results show over **40%** average improvement over baselines in sample efficiency and computational efficiency.

Deep Learning Alpha Pipeline for Cross-Sectional Stock Selection May 2026 – Jun. 2026

Project Independent Quant Research / Deep Learning

- Developed an end-to-end alpha research pipeline on real S&P 500 daily stock data, covering feature engineering, rank-label learning, chronological validation, tuning, backtesting, and robustness evaluation.
- Compared **DLinear, ResMLP, GRU, LSTM, TCN, PatchTST-style Transformer**, and ensemble models for cross-sectional return ranking. PatchTST achieved **25.42%** net top-k long-only return after 5 bps transaction cost, exceeding the **17.25%** buy-and-hold benchmark by **8.17 percentage points**.

Leadership Experience

Student Union, School of Mathematical Sciences, SJTU President Dec. 2024 – Nov. 2025

Honors and Awards

Mathematical Contest in Modeling, Finalist Prize (Top 1%)	2026
China Undergraduate Mathematical Contest in Modeling, National First Prize (Top 0.5%)	2025
Merit Student / Outstanding Student Leader	2024 / 2025
Scholarship of Tung's Foundation / Guo Chenchen Scholarship, SJTU	2024 / 2025

Skills

- Programming: Proficient in Python, C++ and MATLAB; experienced in deep learning model training (Top 1% of SJTU's Kaggle Competitions). Familiar with PyTorch, NumPy, Pandas, and Anaconda.
- Language: Fluent in Chinese (native), English; experienced in academic communication in English.